BCC1313 – ALGORITHMS & COMPLEXITY

Tutorial 5

True/False Questions

1. New items enter a queue at its front.
2. The first item to be added to a queue is the first item to be removed from the queue.
3. If the characters in a string are added to a queue one at a time, characters removed from the queue will occur in the order in which they appear in the original string.
4. A queue can be used to preserve the order of occurrences.
5. The addition of an item to an empty queue is a special case.
6. Rightward drift is a problem that is encountered in a pointer-based implementation of a queue.
7. Operations of the ADT queue can be viewed as general versions of the list and stack operations.
8. In an event-driven simulation of a bank, the departure times of customers are determined by the simulation.
9. In an event-driven simulation of a bank, the arrival of a customer is an internal event.
10. In a time-driven simulation, no action is required between events.

Short Answer Questions

1. What is the meant by the first-in, first-out (FIFO) property of a queue?
2. What is the main difference between a stack and a queue?
3. What is the action performed by the enqueue operation?
4. What is the action performed by the dequeue operation?
5. What is the action performed by the getFront operation?
6. Write the code fragment that inserts a new node into a nonempty queue represented by a linear linked list. Assume that newPtr points to the new node.

7. Write the code fragment that removes the front item from a queue containing more than one item and which represented by a linear linked list.
8. Define the problem of rightward drift in an array-based implementation of a queue.
9. Write the code segment that inserts the item newItem into a queue that is represented by a circular array.
10. What is the advantage an implementation of a queue that uses the ADT list over an implementation that uses a linked list?